Forces Mini-Assessment

TOTAL: 35

_____

1. **State** which word from the list below correctly completes each statement.

Push, motion, Newton, force, spring.

a. All around us, things move or are in _____

b. Whenever there is a change in motion, a _____ has acted.

c. A force can be a _____, a pull or a twist.

d. A force can be measured using a _____ balance.

e. The unit used to measure force is called the _____ .

(5 marks)

2. **List** three things a force can do to an object.

a. _____ _____

b. _____ _____

c. _____ _____

(3 marks)

3. **State** whether the following are true or false.

a. If the forces acting on an object are balanced, then it is not moving.

_____

b. You apply a force when you squeeze a tube of toothpaste.
_____

c. Inertia describes the tendency of an object to change its motion.
_____

d. The less mass an object has, the greater its inertia.
_____ (4 marks)

4. **Describe** an example of a force that:

a. speeds up an object's motion.
_____

_____
_____

_____
_____

b. changes an object's direction.
_____

_____
_____

_____
_____

(2 marks)

5. **Identify** the direction of movement of objects acted upon by these forces:

Force to Left (N)	Force to Right (N)	Forces Upwards (N)	Forces Downwards (N)	Direction of Movement
10	10	10	0	
20	30	0	0	
25	5	15	15	
30	30	10	50	

40	40	100	100	

(5 marks)

6. **Calculate** the average force used to move each box.

Box	Force Required to Move Box			Average
	Trial 1	Trial 2	Trial 3	
1	12	13	11	
2	4.5	5	5	
3	23	26	23.5	
4	11	12.5	10	
5	7	7.3	7.2	

(5 marks)

7. **Classify** each of these as a push, pull or twist force.

   a. Sweeping the floor. _____

   b. Dragging a heavy sports bag along the floor. _____

   c. Throwing a cricket ball. _____

   d. Hitting a golf ball. _____

   e. Tightening wheel nuts on a car. _____

   f. Closing your front door from the inside of the house. _____

   f. Closing your front door from the outside of the house. _____

(7 marks)

8. Forces can be represented by arrows.

   a. **Compare** the forces shown below by stating which:

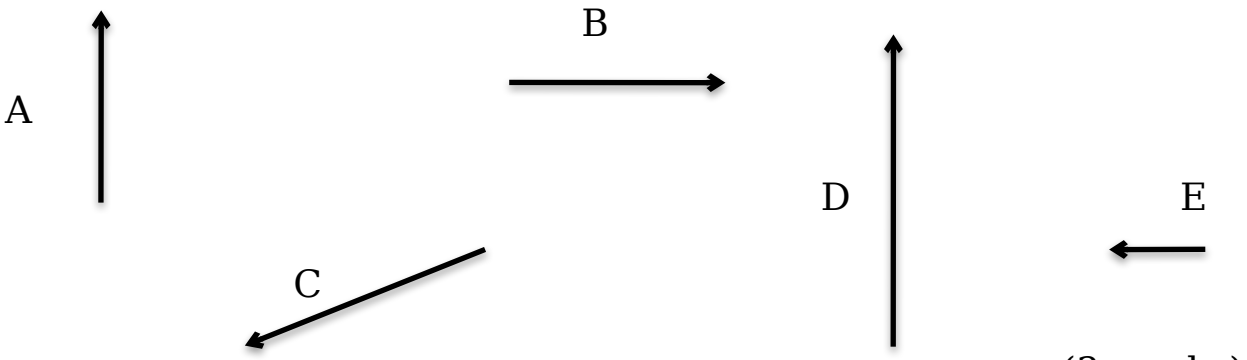
i) force is the largest. _____

ii) two forces are the same.

_____

iii) two forces are acting in the same direction.

_____



(3 marks)

b. If forces B and E acted on an object, **predict** which direction the object would move.

_____

_____

(1 mark)